Hyperspectral Image Processing and Classification in ENVI

1. Introduction to Hyperspectral Data Processing

Hyperspectral imaging provides a high-dimensional data environment that is unparalleled for Earth observation; however, the transition from raw spectral captures to actionable geospatial intelligence requires a rigorous, multi-stage analytical workflow. Because these datasets contain contiguous spectral bands, they are susceptible to high degrees of redundancy and sensor noise. Navigating this complexity requires a structured approach to maintain data integrity and ensure that classification results represent true physical properties.

This report establishes a technical framework consisting of three critical phases: signal optimization via noise reduction, endmember identification through pixel purity analysis, and supervised classification using spectral angles. By following this methodology, analysts can extract precise spectral signatures and "identify the ingredients" of a scene with scientific accuracy. The first priority in this lifecycle is cleaning the raw signal to ensure subsequent algorithms operate on valid data.

2. Phase I: Signal Optimization via Minimum Noise Fraction (MNF) Rotation

The strategic necessity of noise removal is the primary "So What?" factor in hyperspectral analytics. Raw datasets often contain atmospheric interference and sensor-induced noise that can mask subtle spectral features. Minimum Noise Fraction (MNF) rotation acts as a sophisticated data reduction technique that utilizes a noise covariance matrix to decorrelate and rescale noise, essentially performing a two-phase Principal Component Analysis (PCA). By segregating noise from the true spectral signal, MNF rotation significantly increases the reliability and statistical power of the entire analytical pipeline.

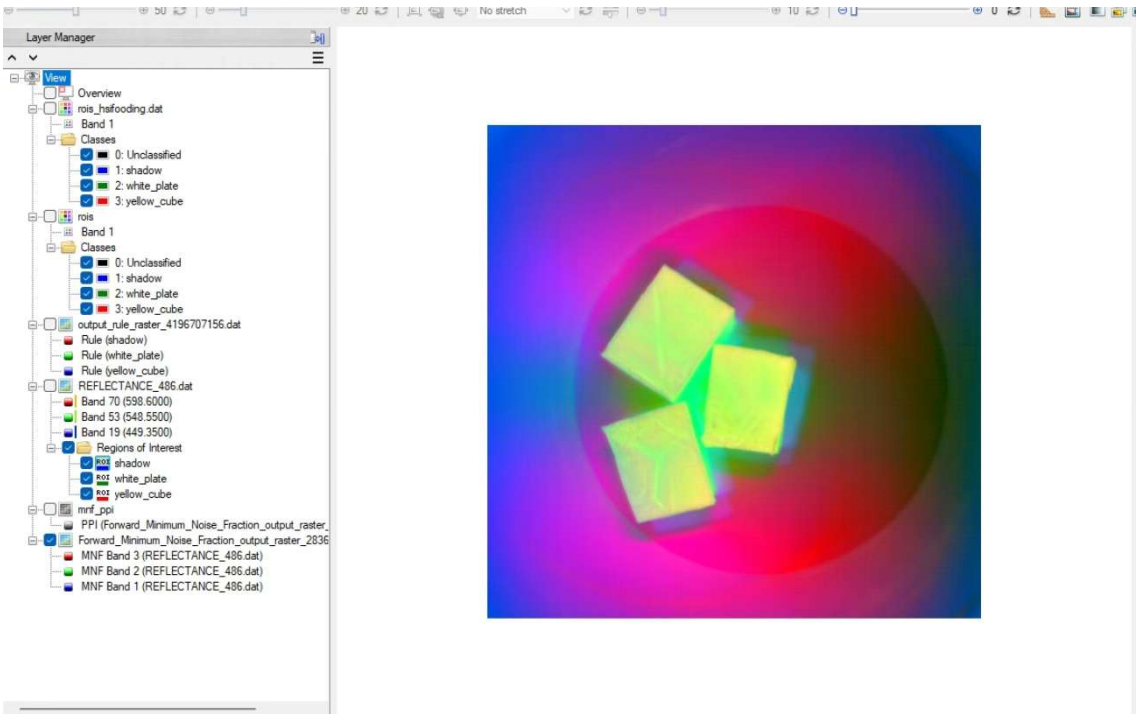
Technical Procedure for MNF Rotation

1. **Initialization:** Open the ENVI environment. Navigate to **File** > **Open** and select the appropriate **.hdr** file for the dataset.
2. **Transformation Path:** In the ENVI Toolbox, navigate to **Transform** > **MNF Rotation** > **Forward MNF Estimate**.
3. **Execution:** Within the window popup, select the **image.dat** file as the input source and click **OK**.

4. **Output Allocation:** Define a specific output destination for the MNF file and click **OK** to execute the rotation.

Technical Note: It is vital to select the image.dat file for the MNF estimate rather than the raw header. This ensures data continuity and maintains the link between spectral bands and their spatial coordinates during the transformation.

This denoised MNF output serves as the mandatory foundation for identifying pure spectral components, as it provides a cleaner feature space for endmember extraction.



3. Phase II: Spectral Feature Extraction using Pixel Purity Index (PPI)

Identifying "pure points," or endmembers, is the cornerstone of accurate hyperspectral mapping. In complex scenes, pixels are often "mixed," containing spectral contributions from multiple materials. The Pixel Purity Index (PPI) is designed to reduce dataset complexity by isolating those rare, spectrally unique pixels that represent a single, unmixed material.

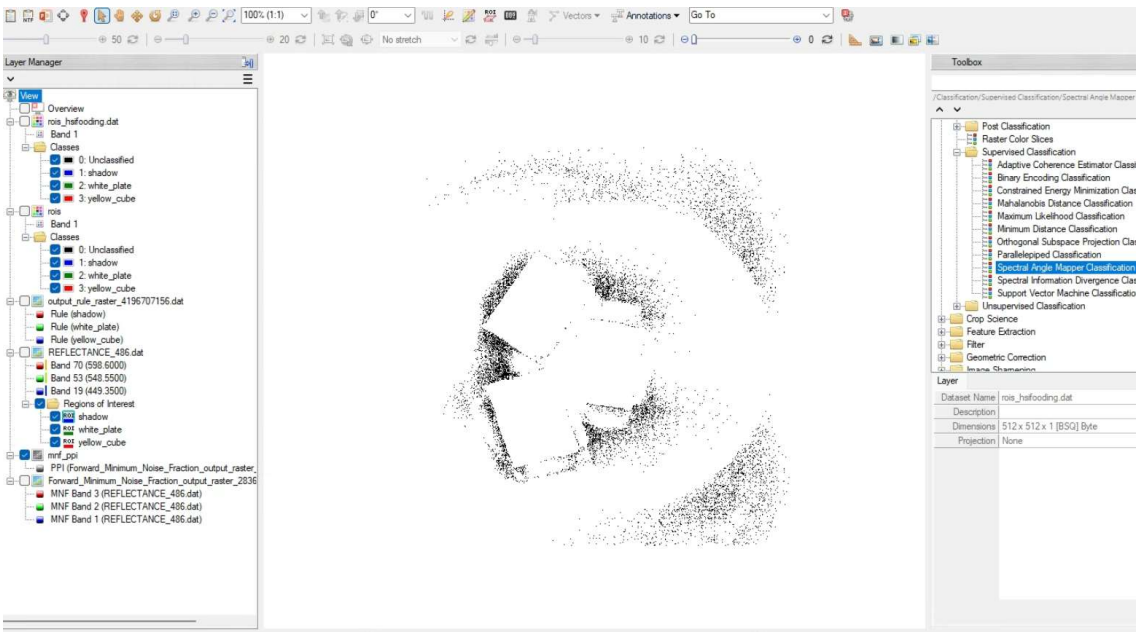
Fast PPI Workflow

To execute the spectral feature extraction, follow this sequence:

- **Navigation:** In the **Toolbox**, navigate to **Spectral Folder** > **Pixel Purity Index** > **Fast PPI**.

- **Parameter Configuration:** Select the **MNF file** generated in Phase I as the input file.
- **Iteration Strategy:** Explicitly set the number of iterations to **5,000**.

The requirement for **5,000 iterations** is critical for statistical robustness. This volume allows the algorithm to repeatedly project the data onto random unit vectors, ensuring the "New Output"—a **PPI Image** where higher pixel values represent higher spectral purity—is stable and representative of the scene's true endmembers. This output allows the analyst to threshold the data and select the most reliable pixels for training.



4. Phase III: Supervised Classification and Spectral Angle Mapping (SAM)

The final phase transitions from data preparation to feature classification. The Spectral Angle Mapper (SAM) is a physically based spectral classification that determines the similarity between an unknown pixel spectrum and a reference spectrum by calculating the "angle" between them in an n-dimensional space (where n is the number of bands). This method is particularly effective as it is relatively insensitive to illumination and albedo effects.

Prerequisite: ROI Development

Before the classification can be executed, the analyst must manually prepare training data. The pure pixels identified in Phase II should serve as the primary guide for placing these regions.

- **Action:** Select the **ROI Tool icon**, located in the **Layer Manager** or the **main toolbar** of the display window.
- **Geometry Selection:** Choose specific geometries (points or polygons) to capture the pure pixels identified during the PPI phase.
- **Labeling Protocol:** It is necessary to **Rename the ROI** for each class (e.g., Vegetation, Water, Soil) to maintain organizational clarity. This process must be repeated until all required classes for the scene have been created.

Final Classification Execution

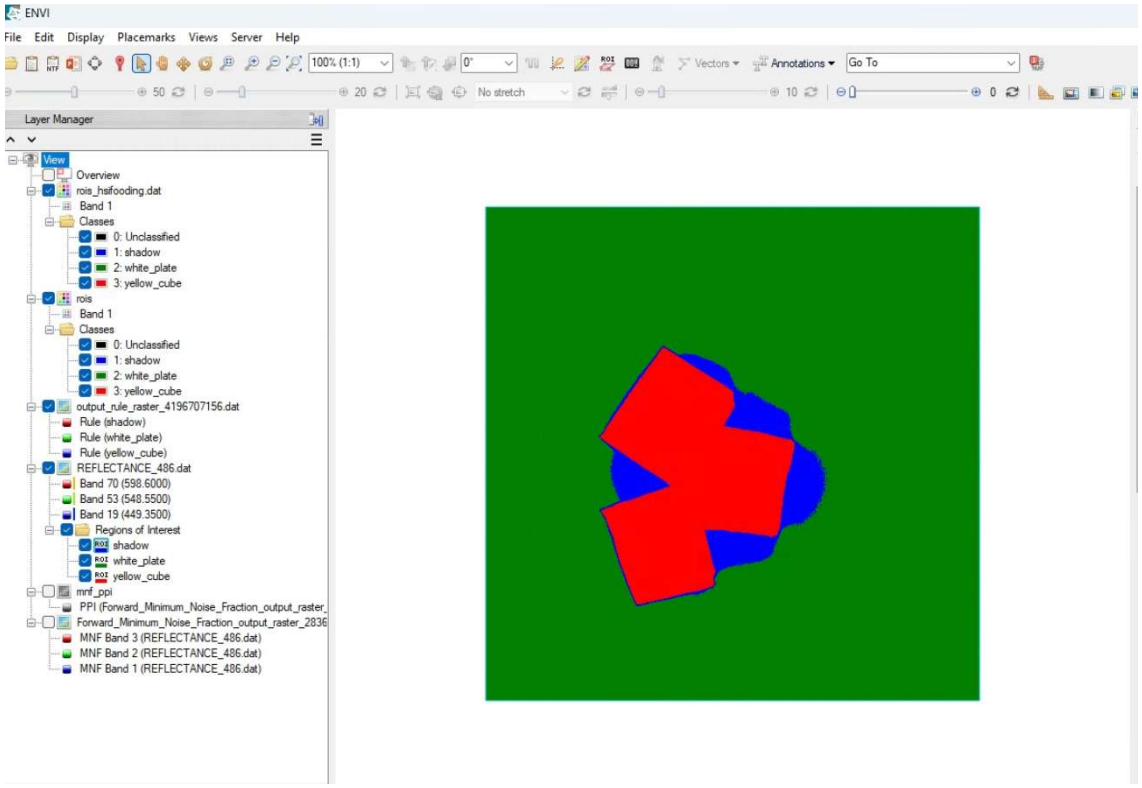
Once the ROIs are established, navigate to the final classification path: **Toolbox** > **Classification** > **Supervised Classification** > **Spectral Angle Mapper**.

By executing this path, the analyst transitions from raw, unorganized pixels to a thematic map that "Identifies the Ingredients" of the scene. The SAM algorithm compares the input data to the renamed ROI references, producing a classified analytical product that categorizes every pixel based on its spectral signature.

5. Procedural Synthesis and Best Practices

The transition from raw data to a classified map relies on the seamless integration of MNF noise reduction, PPI endmember extraction, and SAM classification. This sequence is not merely a collection of tools but a methodology designed to ensure reproducibility and spectral integrity. Skipping noise reduction or utilizing insufficient PPI iterations will directly degrade the accuracy of the final thematic map.

Methodology Checklist

- **Input Verification:** Valid **.hdr** file and associated **image.dat** source.
- **Phase I (Signal):** Forward MNF result used to decorrelate noise.
- **Phase II (Purity):** Fast PPI executed with **5,000 iterations** to produce a **PPI Image**.
- **Phase III (Classification):** A complete set of **renamed ROIs** guided by PPI results, followed by **SAM** execution.

Adhering to this specific sequence ensures that the resulting geospatial intelligence is both precise and scientifically defensible, maintaining the highest standards of hyperspectral analysis.